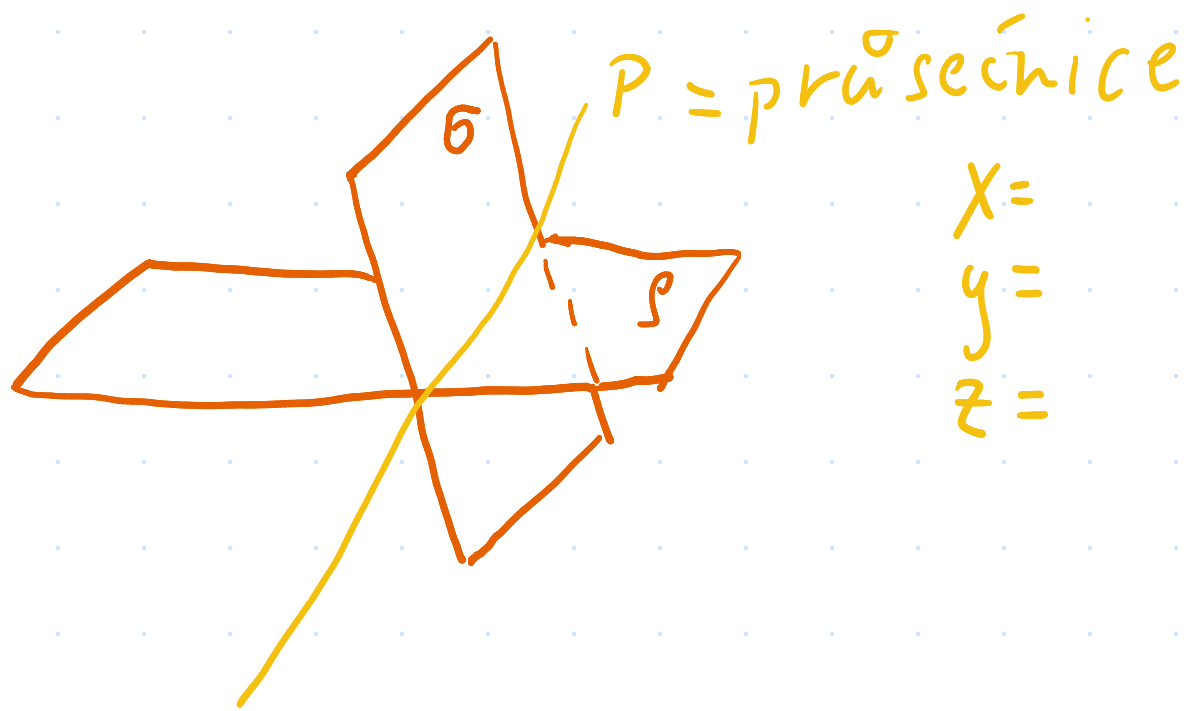


Průsečnice rovin



$$\begin{aligned} 1) \quad & \rho: 2x - y - z - 1 = 0 \\ & \sigma: x + y + 2z - 3 = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} 1) \quad & \rho: 2x - y - z - 1 = 0 \\ & \sigma: x + y + 2z - 3 = 0 \end{aligned}} \right] +$$

$$\begin{aligned} 3x + z - 4 &= 0 \\ z &= 4 - 3x \\ z &= 4 - 3t \end{aligned}$$

$$\begin{aligned} \rho: 2t - y - (4 - 3t) - 1 &= 0 \\ 2t - y - 4 + 3t - 1 &= 0 \\ -y + 5t - 5 &= 0 \\ 5t - 5 &= y \end{aligned}$$

$$\begin{aligned} P: x &= t \\ y &= -5 + 5t \\ z &= 4 - 3t \end{aligned}$$

$$\begin{aligned} 3) \quad & \rho: x - 2y + z + 2 = 0 \\ & \sigma: 2x - y - z - 3 = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} 3) \quad & \rho: x - 2y + z + 2 = 0 \\ & \sigma: 2x - y - z - 3 = 0 \end{aligned}} \right] +$$

$$\begin{aligned} 3x - 3y - 1 &= 0 \\ 3x &= 3y + 1 \\ 3x &= 3t + 1 \\ x &= t + \frac{1}{3} \end{aligned}$$

$$\begin{aligned} 2) \quad & \rho: x + y - z + 3 = 0 \\ & \sigma: 2x - y + z - 9 = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} 2) \quad & \rho: x + y - z + 3 = 0 \\ & \sigma: 2x - y + z - 9 = 0 \end{aligned}} \right] +$$

$$\begin{aligned} 3x - 6 &= 0 \\ x &= 2 \end{aligned}$$

$$z = t$$

$$\begin{aligned} \rho: 2 + y - z + 3 &= 0 \\ y - z + 5 &= 0 \\ y &= z - 5 \\ y &= t - 5 \end{aligned}$$

$$\begin{aligned} P: x &= 2 \\ y &= t - 5 \\ z &= t \end{aligned}$$

$$y = t$$

$$\begin{aligned} \rho: t + \frac{1}{3} - 2t + z + 2 &= 0 \quad / \cdot 3 \\ 3t + 1 - 6t + 3z + 6 &= 0 \\ 3z - 3t + 7 &= 0 \end{aligned}$$

$$\begin{aligned} 3z &= 3t - 7 \\ z &= t - \frac{7}{3} \end{aligned}$$

$$\begin{aligned} x &= t + \frac{1}{3} \\ y &= t \\ z &= t - \frac{7}{3} \end{aligned}$$

Odchylka dvou rovin

$$\cos \varphi = \frac{|\vec{n}_5 \cdot \vec{n}_6|}{|\vec{n}_5| \cdot |\vec{n}_6|} \quad \varphi \in \langle 0^\circ; 90^\circ \rangle$$

$\vec{n}_5 = (n_1, n_2, n_3) \Rightarrow |\vec{n}_5| = \sqrt{n_1^2 + n_2^2 + n_3^2}$

1) $\rho: 2x - y - z - 1 = 0 \quad \vec{n}_\rho = (2, -1, -1)$
 $\sigma: x + y + 2z - 3 = 0 \quad \vec{n}_\sigma = (1, 1, 2)$

$$\cos \varphi = \frac{|2 \cdot 1 - 1 \cdot 1 - 2 \cdot 2|}{\sqrt{6} \cdot \sqrt{6}} = \frac{|-1|}{6} = \frac{1}{6} \Rightarrow \varphi = \underline{\underline{80^\circ}}$$

2) $\rho: x - 2y + 2z = 0 \quad \vec{n}_\rho = (1, -2, 2)$
 $\sigma: x = 1 - t - s$
 $y = t$
 $z = s$
 $\vec{n}_\sigma = (1, 1, 1)$

$$\cos \varphi = \frac{|1 \cdot 1 - 2 \cdot 1 + 2 \cdot 1|}{\sqrt{9} \cdot \sqrt{3}} = \frac{1}{3\sqrt{3}} \Rightarrow \varphi = \underline{\underline{78^\circ}}$$

$$x = 1 - y - z$$

$$0 = -x + 1 - y - z$$

$$0 = x + y + z - 1$$

3) $\rho: 2x - 3y + z - 1 = 0$

$\sigma: x = 2 - s \Rightarrow s = 2 - x$
 $y = 3 - t - s$
 $z = -1 - t \Rightarrow t = -1 - z$

$$y = 3 - (-1 - z) - (2 - x)$$

$$y = 3 + 1 + z - 2 + x$$

$\sigma: 0 = x - y + z + 2$

$$\vec{n}_\rho = (2, -3, 1)$$

$$\vec{n}_\sigma = (1, -1, 1)$$

$$\cos \varphi = \frac{|2 \cdot 1 + 3 \cdot 1 + 1 \cdot 1|}{\sqrt{14} \cdot \sqrt{3}} = \frac{6}{\sqrt{42}} \Rightarrow \varphi = \underline{\underline{22^\circ}}$$